AIMI LAB1 — Pneumonia Classification

412551030 Yih-Ning Huang

I. Introduction

The objective of LAB1 is to classify chest X-ray images as PNEUMONIA or NORMAL (Fig. 1) using residual neural networks. The PNEUMONIA class includes both bacterial and viral pneumonia, which are treated as a single category in this binary classification task. The dataset contains 5,856 images, comprising 1,583 normal images (27.0%) and 4,273 pneumonia images (73.0%). This class imbalance may bias models toward predicting pneumonia, making performance on normal images an important consideration. In this lab, I compared six configurations: ResNet18, ResNet50, and ResNet101, each with and without ImageNet pretrained weights. I evaluated whether pretraining improves convergence and test performance, and whether increasing network depth provides additional benefits under the same training scheme. I hypothesized that pretrained models would converge faster and achieve better test performance than models trained from scratch. However, increasing network depth was not expected to consistently improve generalization, given the limited size of the training dataset. The results showed that ImageNet pretraining improved test accuracy and F1-score across all three architectures. Pretrained ResNet50 achieved the highest test accuracy of 93.11% and pneumonia F1-score of 94.71%. Pretrained models also showed faster initial learning, whereas increasing network depth did not consistently improve test performance.

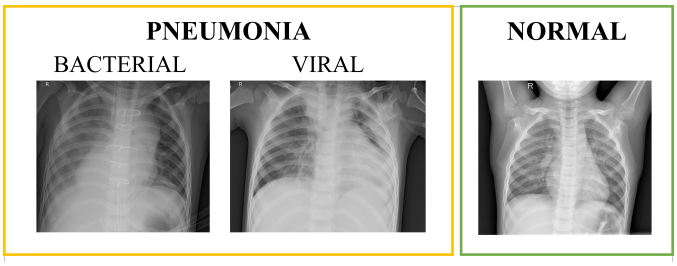


Fig. 1. Representative chest X-ray images showing bacterial pneumonia (left), viral pneumonia (middle), and a normal case (right). Bacterial and viral pneumonia are grouped into the PNEUMONIA class for binary classification.

II. Experimental Setup

A. Model Architectures

Three residual neural network architectures were evaluated: ResNet18, ResNet50, and ResNet101 [1]. The models were implemented in PyTorch using the Torchvision model API (`torchvision.models`). Each architecture was initialized either with ImageNet pretrained weights or with random weights, resulting in six experimental configurations.

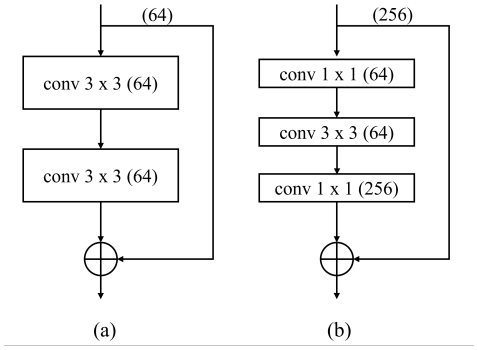


Fig. 2. Residual block structures: (a) a basic block used in ResNet18 and (b) a bottleneck block used in ResNet50 and ResNet101. Numbers in parentheses denote channel counts, and $\oplus$ denotes element-wise addition. Identity shortcuts are shown; batch normalization and ReLU operations are omitted for clarity.

All three architectures use residual connections that add a shortcut to the output of a learned transformation. Fig. 2 illustrates the two block types. ResNet18 uses basic blocks containing two 3×3 convolutions, whereas ResNet50 and ResNet101 use bottleneck blocks containing 1×1, 3×3, and 1×1 convolutions. In the illustrated bottleneck block, the first convolution reduces the channel dimension from 256 to 64, and the final convolution expands it back to 256 before addition to the input. When input and output dimensions differ, a projection shortcut aligns their dimensions before addition.

Table I summarizes the three architectures. ResNet101 differs from ResNet50 primarily in the third residual stage, which contains 23 rather than 6 bottleneck blocks. Following global average pooling, ResNet18 produces a 512-dimensional feature vector, whereas ResNet50 and ResNet101 produce 2048-dimensional feature vectors. For binary chest X-ray classification, the original fully connected classification layer was replaced with a newly initialized linear layer producing two logits corresponding to PNEUMONIA and NORMAL. The residual backbone architecture was retained in all configurations.

B. Data Loading and Preprocessing

1) Dataset Description: The Chest X-Ray Images (Pneumonia) dataset was obtained from Kaggle [2] and originates from the study by Kermany et al. [3]. It contains 5,856 chest X-ray images labeled as NORMAL or PNEUMONIA, with bacterial and viral pneumonia combined into the PNEUMONIA class. The original dataset is divided into 5,216 training images, 16 validation images, and 624 test images. Table II summarizes the class distribution in each partition. The training set

TABLE I
COMPARISON OF RESNET ARCHITECTURES

Component	ResNet18	ResNet50	ResNet101
Network depth	18	50	101
Block type	Basic	Bottleneck	Bottleneck
Conv. layers per block	2	3	3
Initial convolution	7×7, 64 channels, stride 2		
Max pooling	3×3, stride 2		
Stage 1 blocks	2	3	3
Stage 2 blocks	2	4	4
Stage 3 blocks	2	6	23
Stage 4 blocks	2	3	3
Feature dimension	512	2048	2048
Classification outputs	2	2	2

is imbalanced, with substantially more PNEUMONIA images than NORMAL images. In addition, the original validation set contains only eight images per class, making validation performance sensitive to individual predictions.

TABLE II
CLASS DISTRIBUTION IN THE ORIGINAL DATASET

Partition	NORMAL	PNEUMONIA	Total
Training	1,341	3,875	5,216
Validation	8	8	16
Test	234	390	624
Total	1,583	4,273	5,856

2) Image Preprocessing: Grayscale chest X-ray images were converted to three-channel RGB images to match the ResNet input format. All images were resized to 224×224 pixels, converted to tensors with pixel values in $[0, 1]$, and normalized using the ImageNet channel means $(0.485, 0.456, 0.406)$ and standard deviations $(0.229, 0.224, 0.225)$. The same preprocessing was applied to all six model configurations.

3) Data Augmentation: Training images were augmented using random affine transformations, including rotations within $\pm 7°$, horizontal and vertical translations of up to 3%, and scaling factors between 0.95 and 1.05. Newly exposed regions were filled with black pixels. These transformations introduced variations in image orientation, position, and scale during training. No random augmentation was applied to validation or test images.

C. Training Strategy

All training and evaluation were performed using PyTorch on an NVIDIA GeForce RTX 5070 Ti GPU. All six model configurations were trained for 20 epochs using a batch size of 32. All model parameters were optimized using AdamW with an initial learning rate of 1×10^{-4} and a weight decay of 1×10^{-4}. A cosine annealing scheduler gradually reduced the learning rate over the training period.

To address class imbalance, weighted cross-entropy loss was used. Class weights were calculated as $w_c = N/(2N_c)$, where N is the total number of training images and N_c is the number of images in class c. This assigns a larger weight to the less frequent NORMAL class. The weighted per-sample losses were averaged within each mini-batch.

Validation performance was evaluated after every epoch. The checkpoint with the highest validation pneumonia F1-score was selected, with lower validation loss used to break ties. The selected checkpoint was subsequently evaluated on the test set. The same training hyperparameters and checkpoint selection criteria were applied to all six configurations.

D. Evaluation Metrics

Model performance was evaluated using accuracy, F1-score, precision, recall, and specificity. PNEUMONIA was defined as the positive class and NORMAL as the negative class. Accuracy measures overall classification correctness, while F1-score balances pneumonia precision and recall. Recall measures the proportion of pneumonia images correctly identified, and specificity measures the proportion of normal images correctly classified. Confusion matrices were also used to summarize classification errors for each model.

III. EXPERIMENTAL RESULTS

A. Test Performance

Table III summarizes test performance for the six configurations using checkpoints selected by validation performance. Pretrained ResNet50 achieved the highest accuracy (93.11%), F1-score (94.71%), precision (91.02%), and specificity (83.76%). Pretrained ResNet18 and ResNet101 jointly achieved the highest recall (99.49%). Pretraining improved accuracy and F1-score for all three architectures, although increasing network depth did not consistently improve test performance. The corresponding test confusion matrix is shown in Fig. 3. The model correctly classified 196 of 234 normal images and 385 of 390 pneumonia images. There were 38 false positives and 5 false negatives, corresponding to a specificity of 83.76% and a recall of 98.72%.

Fig. 4 presents the training log and final test results of pretrained ResNet50, which achieved the highest test accuracy

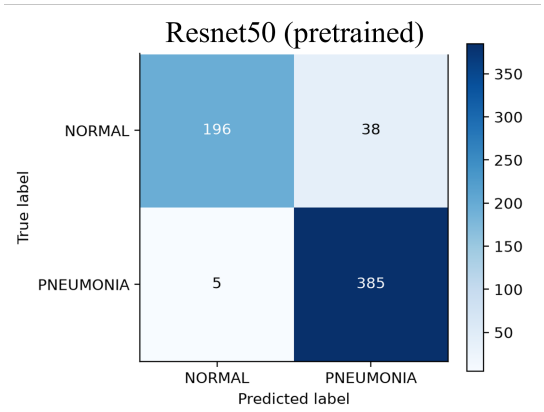


Fig. 3. Test confusion matrix of pretrained ResNet50. Rows represent true labels and columns represent predicted labels. Cell values indicate image counts.

TABLE III
TEST PERFORMANCE OF THE SIX MODEL CONFIGURATIONS

Model	Pretrained	Selected epoch	Accuracy	F1-score	Precision	Recall	Specificity
ResNet18	✓	5	91.03	93.27	87.78	**99.49**	76.92
ResNet50	✓	5	**93.11**	**94.71**	**91.02**	98.72	**83.76**
ResNet101	✓	10	87.34	90.76	83.44	**99.49**	67.09
ResNet18	✗	18	88.14	91.25	84.65	98.97	70.09
ResNet50	✗	11	84.62	88.79	81.55	97.44	63.25
ResNet101	✗	17	87.02	90.30	84.72	96.67	70.94

All performance metrics are reported as percentages. ✓: ImageNet pretrained weights; ✗: random initialization. Bold indicates the highest value and underlining indicates the second-highest distinct value in each metric column across all six configurations. Selected epoch denotes the checkpoint chosen using validation F1-score, with validation loss as the tie-breaker.

```
2026-10-01 15:04:29,377 | INFO | Starting resnet50: pretrained=True
2026-10-01 15:05:39,373 | INFO | resnet50 saved best.pt at epoch 1 (val F1=1.000000, loss=0.184770)
2026-10-01 15:05:39,373 | INFO | resnet50 epoch 01/20 | train loss=0.1171 acc=0.9561 F1=0.9700 | val loss=0.1848 acc=1.0000 F1=1.0000 | lr=0.0001
2026-10-01 15:06:49,664 | INFO | resnet50 saved best.pt at epoch 2 (val F1=1.000000, loss=0.093066)
2026-10-01 15:06:49,665 | INFO | resnet50 epoch 02/20 | train loss=0.0615 acc=0.9797 F1=0.9862 | val loss=0.0931 acc=1.0000 F1=1.0000 | lr=9.93844e-05
2026-10-01 15:07:56,631 | INFO | resnet50 epoch 03/20 | train loss=0.0454 acc=0.9831 F1=0.9886 | val loss=1.6614 acc=0.6875 F1=0.7619 | lr=9.75528e-05
2026-10-01 15:09:04,122 | INFO | resnet50 epoch 04/20 | train loss=0.0311 acc=0.9891 F1=0.9926 | val loss=0.3501 acc=0.9375 F1=0.9412 | lr=9.45503e-05
2026-10-01 15:10:15,190 | INFO | resnet50 saved best.pt at epoch 5 (val F1=1.000000, loss=0.005547)
2026-10-01 15:10:15,190 | INFO | resnet50 epoch 05/20 | train loss=0.0260 acc=0.9904 F1=0.9935 | val loss=0.0055 acc=1.0000 F1=1.0000 | lr=9.04508e-05
2026-10-01 15:11:22,159 | INFO | resnet50 epoch 06/20 | train loss=0.0204 acc=0.9931 F1=0.9953 | val loss=0.0210 acc=1.0000 F1=1.0000 | lr=8.53553e-05
2026-10-01 15:12:29,053 | INFO | resnet50 epoch 07/20 | train loss=0.0136 acc=0.9944 F1=0.9962 | val loss=0.1160 acc=1.0000 F1=1.0000 | lr=7.93893e-05
2026-10-01 15:13:36,270 | INFO | resnet50 epoch 08/20 | train loss=0.0054 acc=0.9979 F1=0.9986 | val loss=0.0836 acc=1.0000 F1=1.0000 | lr=7.26995e-05
2026-10-01 15:14:41,603 | INFO | resnet50 epoch 09/20 | train loss=0.0094 acc=0.9977 F1=0.9985 | val loss=0.1475 acc=1.0000 F1=1.0000 | lr=6.54508e-05
2026-10-01 15:15:45,350 | INFO | resnet50 epoch 10/20 | train loss=0.0087 acc=0.9967 F1=0.9978 | val loss=0.1654 acc=1.0000 F1=1.0000 | lr=5.78217e-05
2026-10-01 15:16:51,271 | INFO | resnet50 epoch 11/20 | train loss=0.0039 acc=0.9988 F1=0.9992 | val loss=0.0074 acc=1.0000 F1=1.0000 | lr=5e-05
2026-10-01 15:17:59,472 | INFO | resnet50 epoch 12/20 | train loss=0.0048 acc=0.9981 F1=0.9987 | val loss=0.1564 acc=1.0000 F1=1.0000 | lr=4.21783e-05
2026-10-01 15:19:06,692 | INFO | resnet50 epoch 13/20 | train loss=0.0021 acc=0.9992 F1=0.9995 | val loss=0.0374 acc=1.0000 F1=1.0000 | lr=3.45492e-05
2026-10-01 15:20:11,060 | INFO | resnet50 epoch 14/20 | train loss=0.0013 acc=0.9996 F1=0.9997 | val loss=0.0359 acc=1.0000 F1=1.0000 | lr=2.73005e-05
2026-10-01 15:21:12,100 | INFO | resnet50 epoch 15/20 | train loss=0.0008 acc=0.9998 F1=0.9999 | val loss=0.0634 acc=1.0000 F1=1.0000 | lr=2.06107e-05
2026-10-01 15:22:16,366 | INFO | resnet50 epoch 16/20 | train loss=0.0008 acc=0.9998 F1=0.9999 | val loss=0.0199 acc=1.0000 F1=1.0000 | lr=1.46447e-05
2026-10-01 15:23:24,793 | INFO | resnet50 epoch 17/20 | train loss=0.0005 acc=0.9998 F1=0.9999 | val loss=0.0269 acc=1.0000 F1=1.0000 | lr=9.54915e-06
2026-10-01 15:24:34,690 | INFO | resnet50 epoch 18/20 | train loss=0.0006 acc=1.0000 F1=1.0000 | val loss=0.0143 acc=1.0000 F1=1.0000 | lr=5.44967e-06
2026-10-01 15:25:42,673 | INFO | resnet50 epoch 19/20 | train loss=0.0005 acc=0.9998 F1=0.9999 | val loss=0.0313 acc=1.0000 F1=1.0000 | lr=2.44717e-06
2026-10-01 15:26:50,558 | INFO | resnet50 epoch 20/20 | train loss=0.0004 acc=1.0000 F1=1.0000 | val loss=0.0226 acc=1.0000 F1=1.0000 | lr=6.15583e-07
2026-10-01 15:26:57,845 | INFO | FINAL TEST: {"accuracy": 0.9310897435897436, "f1": 0.947109471094711, "precision": 0.9101654846335697, "recall":
0.9871794871794872, "specificity": 0.8376068376068376, "confusion_matrix": [[196, 38], [5, 385]], "model": "resnet50", "best_epoch": 5}
2026-10-01 15:26:58,105 | INFO | Saved results to I:\NYCUCS\AIMI\lab1\outputs_pretrained
```

Fig. 4. Training log and final test results of pretrained ResNet50. The checkpoint selected at epoch 5 achieved the highest test accuracy and F1-score among the six configurations.

and F1-score among the six configurations. The checkpoint selected at epoch 5 achieved a test accuracy of 93.11% and a pneumonia F1-score of 94.71%.

B. Training and Validation Curves

Fig. 5 compares training and validation performance over 20 epochs. Pretrained models achieved higher training accuracy and F1-score during the early epochs than models trained from scratch, indicating faster initial learning. By the final epoch, all configurations achieved high training performance.

Validation performance fluctuated more strongly, particularly for models trained from scratch. Pretrained ResNet50 maintained validation accuracy and F1-score of 1.0 from epoch 5 onward. However, the validation set contained only 16 images, so these perfect scores should be interpreted cautiously. A single misclassified image changes validation accuracy by 6.25 percentage points.

IV. DISCUSSION

This experiment showed that ImageNet pretraining improved test performance and accelerated initial learning, whereas greater network depth did not consistently provide additional benefits. Pretrained ResNet50 achieved the best overall performance among the six configurations. However, the small validation set and peripheral activations observed in selected Grad-CAM examples highlight limitations in checkpoint selection and the interpretation of model predictions.

A. Network Depth and ImageNet Pretraining

ImageNet pretraining improved test accuracy and pneumonia F1-score for all three architectures. The accuracy gains were 2.88, 8.49, and 0.32 percentage points for ResNet18, ResNet50, and ResNet101, respectively. Pretrained ResNet50 achieved the best overall performance, with an accuracy of 93.11% and an F1-score of 94.71%.

The training curves also showed faster initial learning with pretraining. As shown in Table III, the selected checkpoints occurred at epochs 5, 5, and 10 for the pretrained models, compared with epochs 18, 11, and 17 for their randomly initialized models. These earlier selected epochs are consistent with the benefits of pretraining, but do not independently establish convergence because checkpoint selection also depends on validation variability.

On the other hand, greater network depth did not consistently improve test performance. Although ResNet101 contains more bottleneck blocks than ResNet50, pretrained ResNet101 performed worse on both accuracy and F1-score. This suggests that additional depth was not necessarily ben-

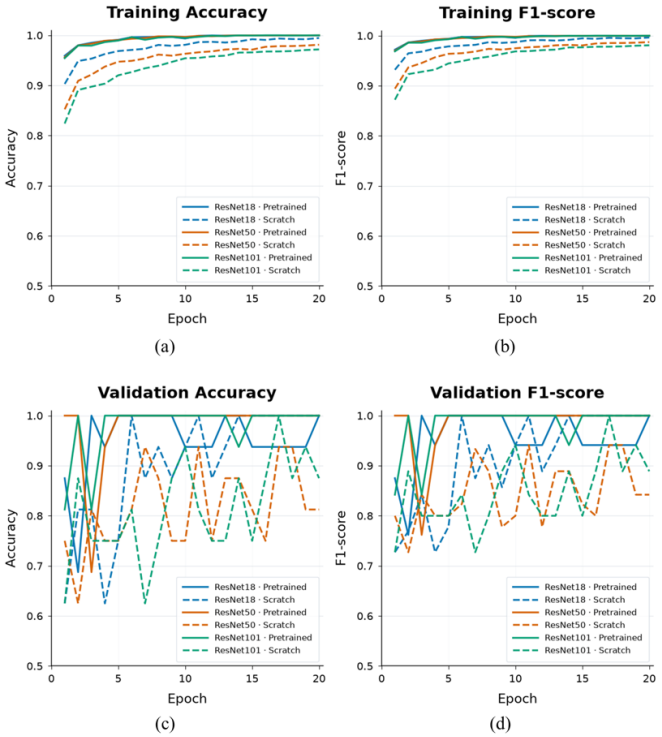


Fig. 5. Learning curves for the six configurations: (a) training accuracy, (b) training pneumonia F1-score, (c) validation accuracy, and (d) validation pneumonia F1-score. Colors distinguish ResNet18, ResNet50, and ResNet101; solid lines indicate ImageNet pretraining and dashed lines indicate training from scratch.

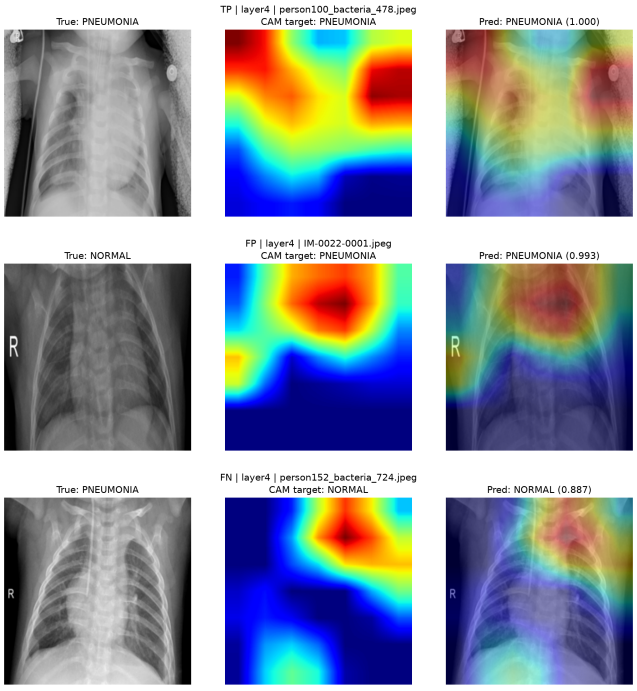


Fig. 6. Selected Grad-CAM examples from `layer4`: a true positive (top), a false positive (middle), and a false negative (bottom). Each row shows the original image, activation map, and overlay. Maps target the predicted class: PNEUMONIA for the top and middle rows, and NORMAL for the bottom row. Warmer colors indicate stronger positive contributions to the target-class score within each map.

eficial under the shared 20-epoch training schedule. Limited training data, optimization differences, and overfitting are possible explanations, but the present experiment cannot distinguish their individual contributions.

B. Validation Reliability

The original validation set contained only 16 images. Consequently, one incorrect prediction changes validation accuracy by 6.25 percentage points. Perfect validation scores therefore provide limited evidence of generalization, and selecting checkpoints using this small set may be unstable. A larger validation set with group-based partitioning would provide a more reliable comparison.

C. Grad-CAM Interpretation

To interpret the model decision-making process, Grad-CAM [4] was applied to pretrained ResNet50 using feature maps from `layer4`. The method uses target-class gradients to weight feature maps and generate a coarse localization map. Each visualization targets the predicted class: PNEUMONIA for true-positive and false-positive cases, and NORMAL for false-negative cases.

As shown in Fig. 6, the true-positive example exhibits strong activation near the shoulder and image boundary, suggesting possible reliance on non-pulmonary cues. The false-positive case has a PNEUMONIA probability of 99.30%, whereas the false-negative case has a NORMAL probability of 88.70%, illustrating that high predicted probabilities can accompany incorrect classifications.

These maps raise concerns about whether the model relies on pulmonary abnormalities or peripheral image cues. However, saliency maps alone cannot verify lesion localization: Saporta et al. [5] found that Grad-CAM localization remained substantially below the radiologist benchmark. Recent approaches such as ChEX [6] incorporate anatomical and pathology bounding-box supervision to support explicit localization and region-level interpretation. Future work could combine lung-region masking with lesion-level supervision and evaluate localization against expert annotations. Lung masking may restrict peripheral cues, but does not by itself ensure that the model identifies pneumonia lesions.

V. GitHub Repository

The source code, installation instructions, and commands for reproducing the experiments are available at: https://github.com/yihmnz/AIMI2026Fall-LAB1.

References

[1] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

[2] P. Mooney, "Chest X-Ray images (pneumonia)," Kaggle. [Online]. Available: https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia

[3] D. S. Kermany, M. Goldbaum, W. Cai, C. C. S. Valentim, H. Liang, S. L. Baxter, A. McKeown, G. Yang, X. Wu, F. Yan *et al.*, "Identifying medical diagnoses and treatable diseases by image-based deep learning," *Cell*, vol. 172, no. 5, pp. 1122–1131.e9, 2018.

[4] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 618–626.

[5] A. Saporta, X. Gui, A. Agrawal, A. Pareek *et al.*, "Benchmarking saliency methods for chest X-ray interpretation," *Nature Machine Intelligence*, vol. 4, pp. 867–878, 2022.

[6] P. Müller, G. Kaissis, and D. Rueckert, "ChEX: Interactive localization and region description in chest X-rays," in *European Conference on Computer Vision (ECCV)*, 2024. [Online]. Available: https://www.ecva.net/papers/eccv_2024/papers_ECCV/papers/03114.pdf